Session 13: Excel Formulas: SUM, AVERAGE, MIN, MAX

Presented by

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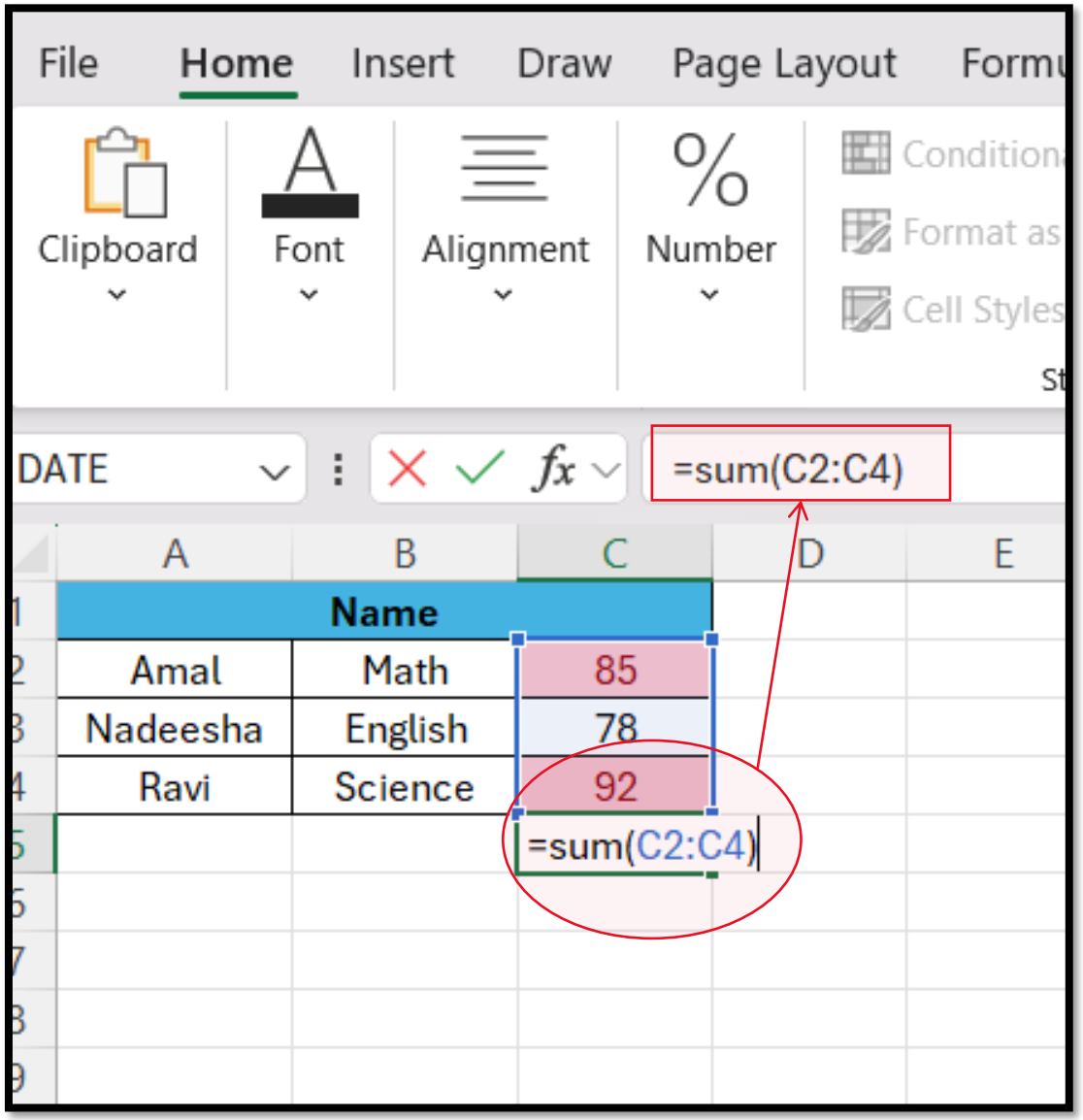
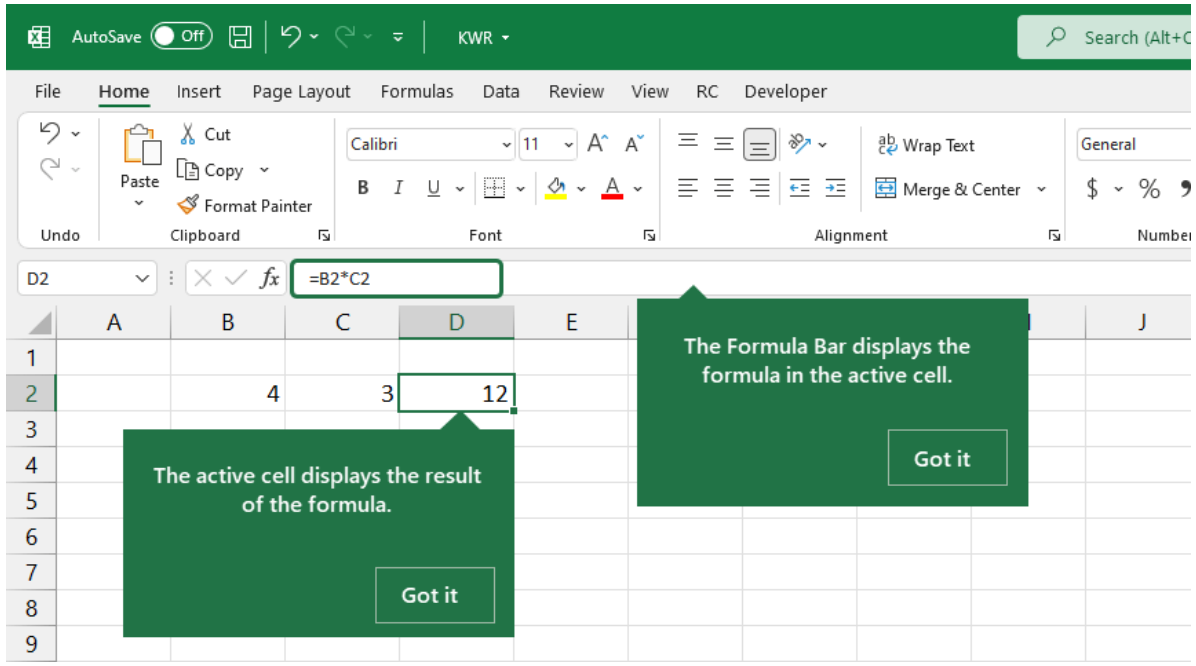
Why Use Formulas in Excel?

- *Formulas allow automatic calculations.*
- *Save time and reduce manual errors.*
- *Useful for totals, averages, finding min/max.*
- *Always start with = sign.*



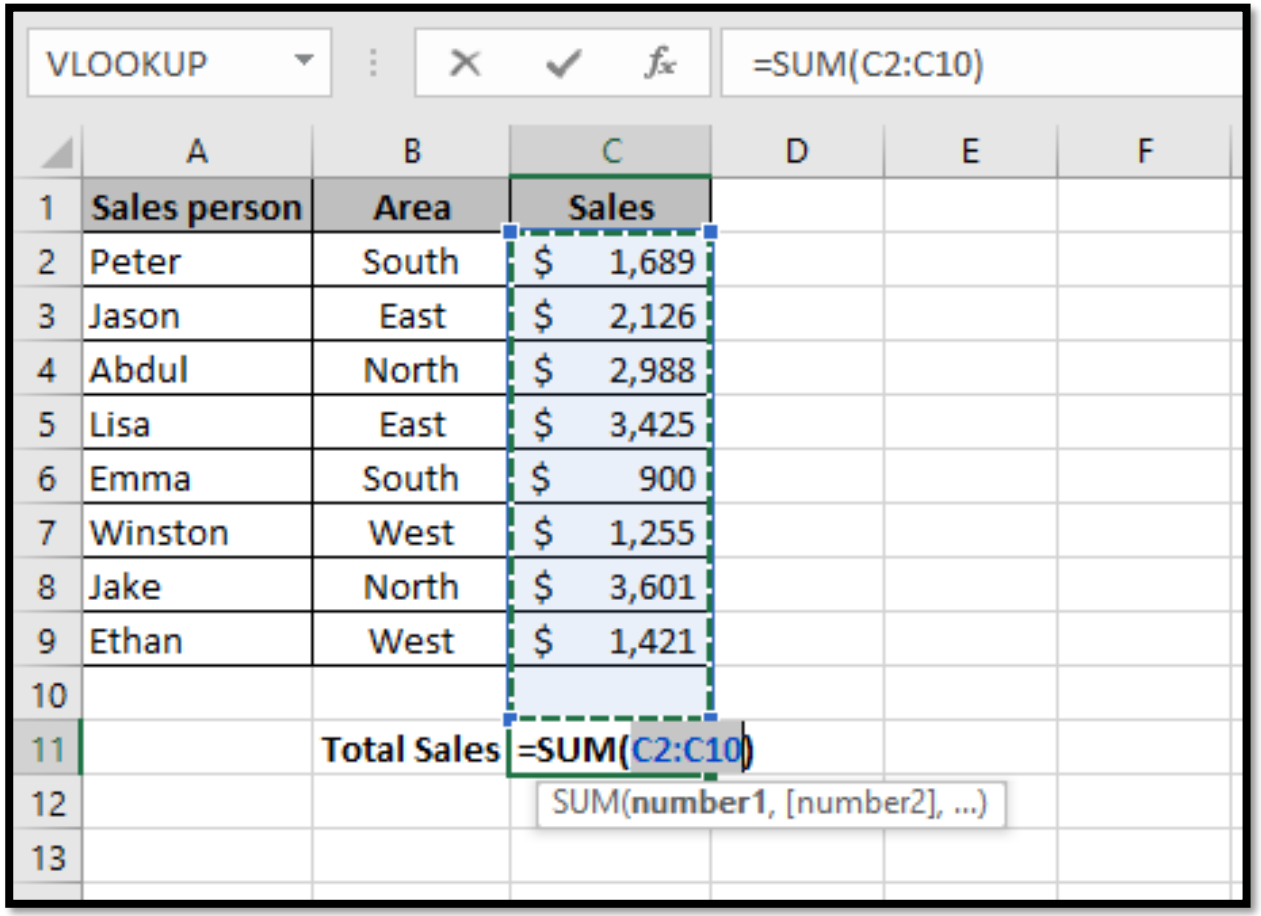
Understanding Formulas



- *Formulas start with =.*
- *You can add, subtract, multiply, divide.*
- *Example: =A1 + B1 adds values in A1 and B1.*

The SUM Function

- *Adds a range of numbers quickly.*
- *Syntax: =SUM(A1:A5)*
- *Example: Total marks of students.*



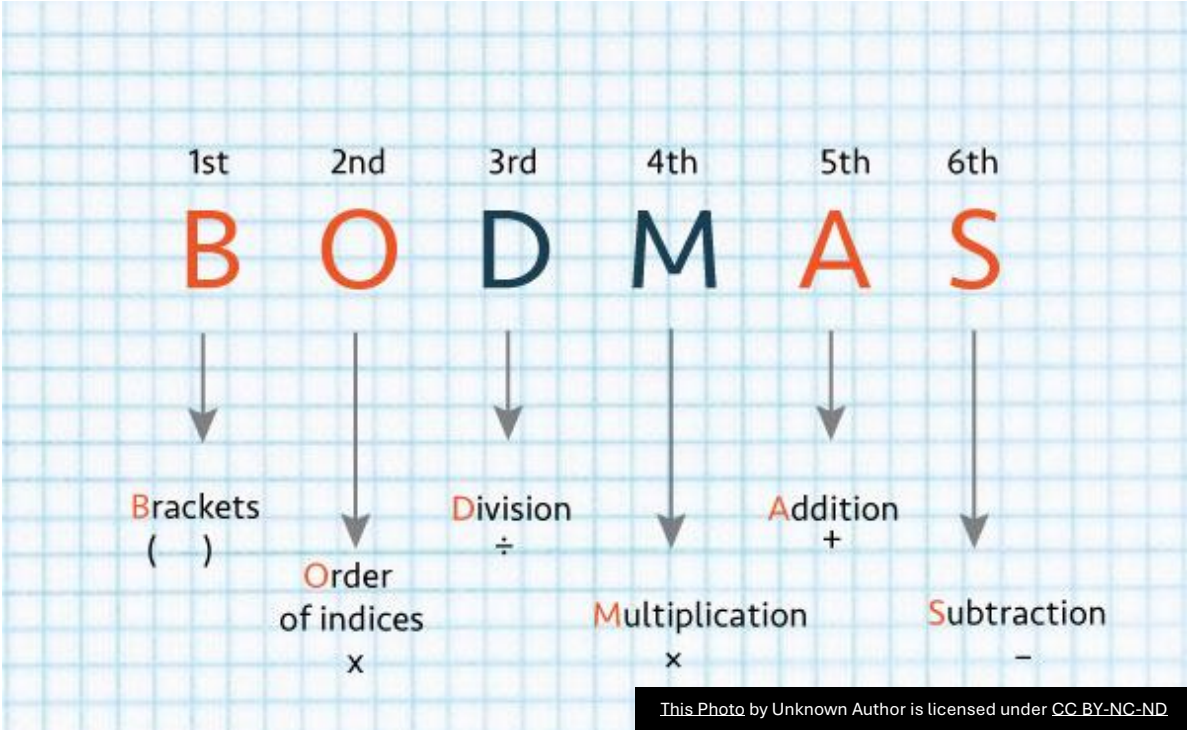
The AVERAGE Function

- *Finds the mean (average) of a range of numbers.*
- *Syntax: =AVERAGE(A1:A5)*
- *Example: Average marks in a class.*

	A	B	C	D	E	F
1	Sl. No.	Items	Quantity	Price (USD)		
2	1	Item 1	32	100		
3	2	Item 2	30	200		
4	3	Item 3	28	500		
5	4	Item 4	27	250		
6	5	Item 5	26	10		
7	6	Item 6	30	60		
8	7	Item 7	31	70		
9	8	Item 8	35	20		
10	9	Item 9	40	40		
11	10	Item 10	42	77		
12	11	Item 11	44	30		
13	12	Item 12	35	29		
14	13	Item 13	37	90		
15	14	Item 14	38	120		
16				=AVERAGE(D2:D15)		
17						
18						

Details

How Excel Thinks: BODMAS



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- Excel does not calculate left-to-right. It follows math rules!
- Example: =5+2*3 gives 11 (not 21).
- Tip: Always use brackets to control the math: =(5+2)*3 gives 21.

The MIN Function

- *Finds the smallest value in a range.*
- *Syntax: =MIN(A1:A5)*
- *Example: Lowest mark in a class.*

C2 =MAX(MIN(B2,B16),B15)

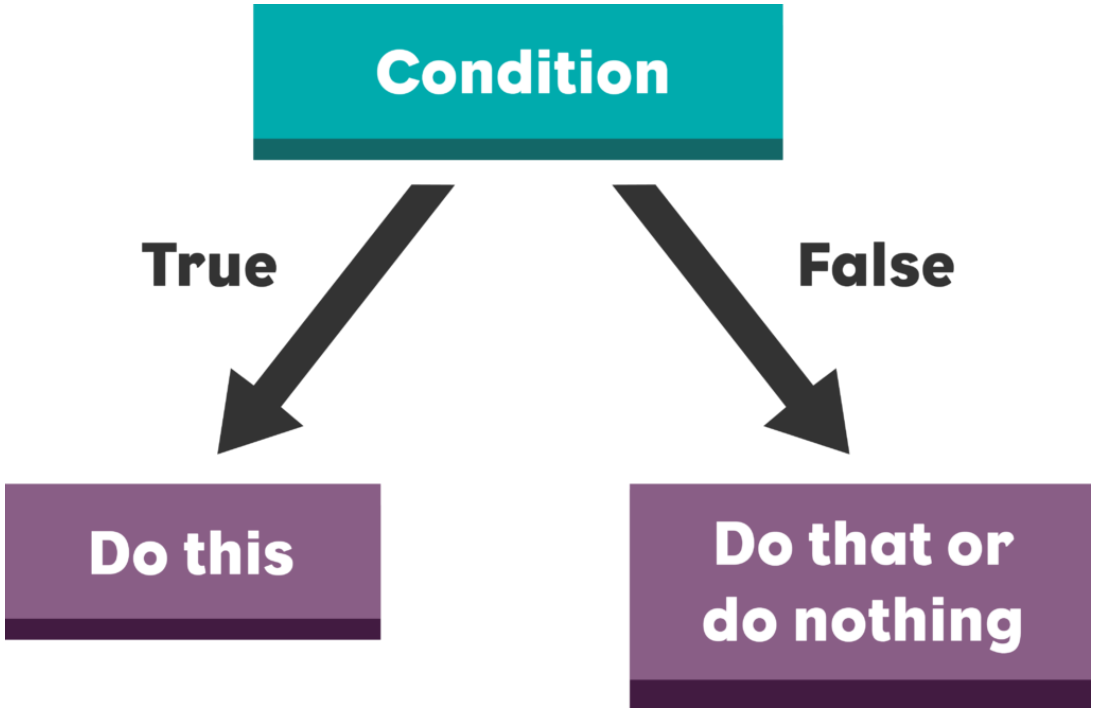
	A	B	C	D	E
1	**Student**	**Score**	**Corrected Scores**		
2	Andy	68	68		
3	Bob	88	88		
4	Chad	94	94		
5	Doug	101	100		
6	Eric	113	100		
7	Frank	78	78		
8	Greg	80	80		
9	Henry	-85	0		
10	Isaac	79	79		
11	John	93	93		
12	Kendall	92	92		
13	Luke	90	90		
14					
15	**Min**	0			
16	**Max**	100			
17					

Making Decisions with the IF Function

- The IF function checks if a condition is true or false.
- Structure:

=IF(Logical_Test, Value_if_True, Value_if_False)

- Example for Marks: * =IF(B2>=50, "Pass", "Fail")
- If the student gets 50 or more, Excel writes "Pass". Otherwise, it writes "Fail".



The MAX Function

- *Finds the largest value in a range.*
- *Syntax: =MAX(A1:A5)*
- *Example: Highest mark in a class.*

C2 =MAX(MIN(B2,B16),B15)

	A	B	C	D	E
1	**Student**	**Score**	**Corrected Scores**		
2	Andy	68	68		
3	Bob	88	88		
4	Chad	94	94		
5	Doug	101	100		
6	Eric	113	100		
7	Frank	78	78		
8	Greg	80	80		
9	Henry	-85	0		
10	Isaac	79	79		
11	John	93	93		
12	Kendall	92	92		
13	Luke	90	90		
14					
15	**Min**	0			
16	**Max**	100			
17					

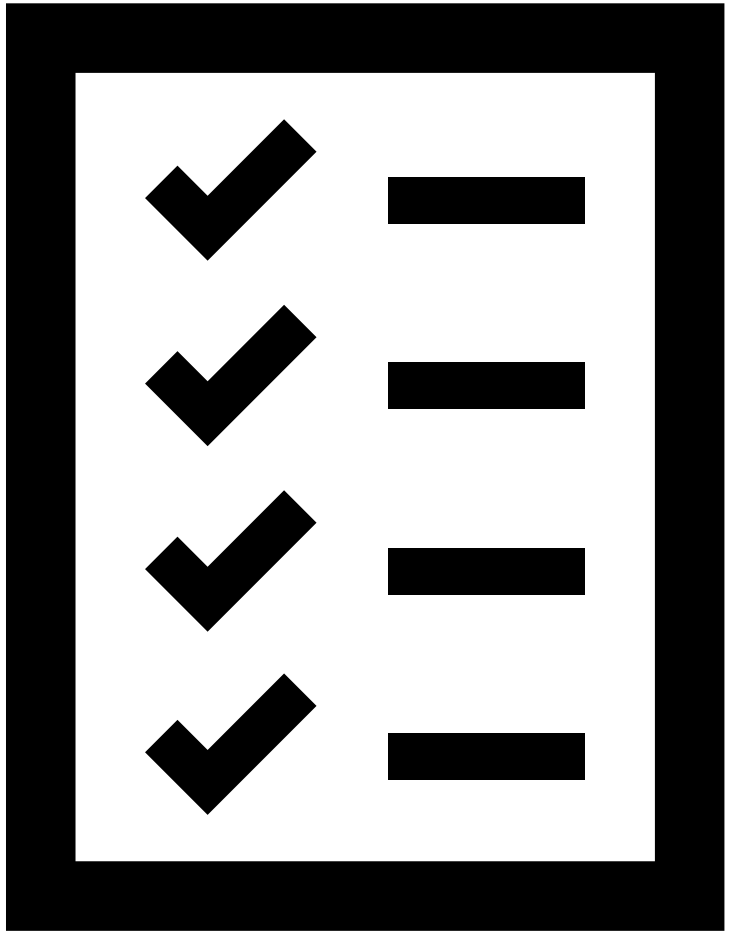
Practical Example Table

✅ *Practice:*

- *Total (SUM)*
- *Average (AVERAGE)*
- *Lowest (MIN)*
- *Highest (MAX)*

Name	Marks
Amal	85
Nadeesha	78
Ravi	92
Sahan	70
Dilani	88
Total (SUM)	
Average (AVERAGE)	
Lowest (MIN)	
Highest (MAX)	

Practice Task



1. *Create the above table.*
2. *Apply formulas below the table.*
3. *Use the IF function to add a 'Status' column (Pass if >= 50, Fail if < 50)*
4. *Format: Bold headers, borders, light header color.*
5. *Save*

Recap & Next Session

You learned:

- *Writing formulas*
- *Using SUM, AVERAGE, MIN, MAX*
- *Applying formulas practically*

Next: Sorting Data & Working with Tables in Excel.

Pro Tip: Data Grouping

What it does: Allows you to collapse/expand rows or columns using + and - buttons.

When to use it: To hide detailed data (like specific expenses) and only show totals.

How to do it:

1. Select the rows you want to hide (e.g., Rows 5, 6, and 7 containing individual expenses).
2. Go to the **Data** tab on the Ribbon.
3. Click **Group** (on the far right, under Outline).
4. Click the - button on the left margin to collapse them!

Send Me Your Thoughts (Anonymously!)

Scan the QR code and share your questions or feedback

https://forms.gle/4WwKrDtcEDtG3X1BA

Thank You!